VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belgaum-590018

IDP Project Report

on

"AI BASED VIRTUAL TRY- ON SYSTEM FOR ONLINE CLOTHING SHOPPING "

Submitted in partial fulfillment of the requirements for the
Second Semester of the Bachelor of Engineering Degree, towards the completion of the
Interdisciplinary Project based learning ,
Department of Basic Sciences.

by

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DEPARTMENT OF BASIC SCIENCES

CERTIFICATE

This is to certify that the File Structures IDP project entitled "AI Based Virtual Try -On System for Online Clothing Shopping" has been successfully carried out by Siddharth R (1CR25IS162), Sinchana S Sooda (1CR25IS163) , Bindhushree .G (1CR25AD030) and Bhawesh Mishra (1CR25AD029) bonafide students of **CMR Institute of Technology.**

The project is submitted in partial fulfillment of the requirements for the First Semester of the Bachelor of Engineering Degree, towards the completion of the **Interdisciplinary Project based learning** , **Department of Basic Sciences.**

It is further certified that all corrections and suggestions indicated during the Internal Assessment have been duly incorporated in the project report submitted to the departmental library. This File Structures IDP project report has been reviewed and approved as it satisfies the academic requirements prescribed for the said degree.

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Signature of Guide	**Signature of HOD**
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External Viva

Name of the examiners   Signature with date

1.

ACKNOWLEDGEMENT

I sincerely express my gratitude to **Dr. Sanjay Jain**, Principal, CMR Institute of Technology, Bangalore, for providing a supportive academic environment.

I extend my thanks to **Dr. Fazlur Rahaman**, **HoD, Dept. of Chemistry, CMRIT** for his valuable guidance and support.

I am especially grateful to my internal guide, Assistant Professor, Chithra M, Department of EEE, for her constant encouragement and guidance throughout this project.

I also thank all the faculty members, non-teaching staff, and others who contributed directly or indirectly to the successful completion of this work.

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Abstract

The AI-Based Virtual Try-On System for Online Clothing Shopping is an innovative web-based platform designed to improve the online fashion shopping experience. Traditional e-commerce platforms often fail to provide customers with a clear understanding of how clothes will look before purchase, leading to uncertainty and increased product return rates. The proposed system addresses this issue by integrating Artificial Intelligence (AI) and API-based virtual try-on technologies to create an interactive and realistic shopping experience.

The system allows users to upload their image and virtually preview different outfits using AI-powered image processing APIs. The selected clothing items are digitally integrated onto the user's image to generate a virtual try-on preview. The platform also includes a responsive and modern frontend inspired by popular fashion shopping websites, enabling users to browse products, explore categories, manage shopping carts, and interact with the platform smoothly.

The project is developed using HTML, CSS, and JavaScript for frontend implementation, while AI APIs support virtual try-on functionality and image processing features. The system enhances customer confidence, improves user engagement, and reduces product return rates caused by poor visualization or fitting uncertainty. This project demonstrates how AI integration and smart web technologies can transform the future of online fashion retail by creating a more personalized, intelligent, and interactive shopping experience.

Keywords:

Artificial Intelligence, Virtual Try-On, API Integration, Online Clothing Shopping, E-Commerce, Image Processing, Responsive Web Design, Fashion Technology, AI APIs, Smart Shopping System.

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Chapter 1

INTRODUCTION

Online shopping has transformed the fashion industry by allowing customers to browse and purchase clothing products from anywhere. However, one of the major limitations of online clothing shopping is the inability of customers to physically try clothes before purchasing. This often leads to uncertainty regarding fit, appearance, and style compatibility, resulting in customer dissatisfaction and high product return rates.

The objective of this project is to develop an AI Based Virtual Try-On System for Online Clothing Shopping using Computer Vision. The system enables users to virtually try outfits using uploaded images or live camera input. By integrating Artificial Intelligence (AI), image processing, and Computer Vision techniques, the platform provides a realistic visualization of clothing items on the user's body.

The proposed system improves user confidence, enhances shopping convenience, and creates a more interactive and personalized online shopping experience.

1.1 Brief History of Online Shopping Systems

Traditional online shopping websites initially provided only static images of clothing products. Customers had to rely on size charts and imagination to understand how outfits would look on them. As e-commerce platforms evolved, features like zoom previews, 360-degree product views, and AI recommendations were introduced to improve user experience.

With advancements in Artificial Intelligence and Computer Vision, virtual try-on systems emerged as an innovative solution. These systems use image recognition, body detection, and augmented visualization techniques to digitally overlay clothing items onto users, making online fashion shopping more realistic and interactive.

1.2 Modern AI-Based Virtual Try-On System

Modern virtual try-on systems use technologies such as AI, Machine Learning, Computer Vision, and Augmented Reality (AR) to provide realistic clothing previews. These systems detect body posture, body shape, and facial alignment to accurately fit digital clothing onto users.

Features of modern virtual try-on systems include:

- Real-time clothing visualization
- AI-based body detection and pose estimation
- Personalized fashion recommendations
- Live camera integration
- Interactive shopping interfaces
- Reduced product return rates

The proposed project combines a modern frontend interface with AI-powered visualization features to create an advanced online fashion shopping platform.

Chapter 2

Problem Statement

2.1 Description

Online clothing shoppers often face difficulties in selecting suitable outfits because they cannot physically try clothes before purchasing. Product images alone are insufficient to determine how clothes will fit or appear on different body types. This creates confusion, lack of confidence, and increased chances of returning purchased items.

Existing solutions either lack realistic visualization or require expensive hardware and complex setups. Therefore, there is a need for an intelligent, user-friendly, and cost-effective virtual try-on system that can provide accurate clothing previews for users in real time.

2.2 Challenge Statement

The major challenges addressed in this project include:

- Providing realistic virtual clothing visualization
- Accurate body detection and alignment
- Supporting different body shapes and poses
- Reducing clothing mismatch issues
- Creating a responsive and interactive frontend interface
- Integrating AI and Computer Vision efficiently
- Ensuring smooth performance across devices

The project aims to solve these challenges by developing an AI-driven virtual try-on platform for online fashion shopping.

Chapter 3

3.1 Design Thinking Process

a) Empathize

User surveys identified problems related to fitting uncertainty and lack of realistic outfit visualization in online shopping.

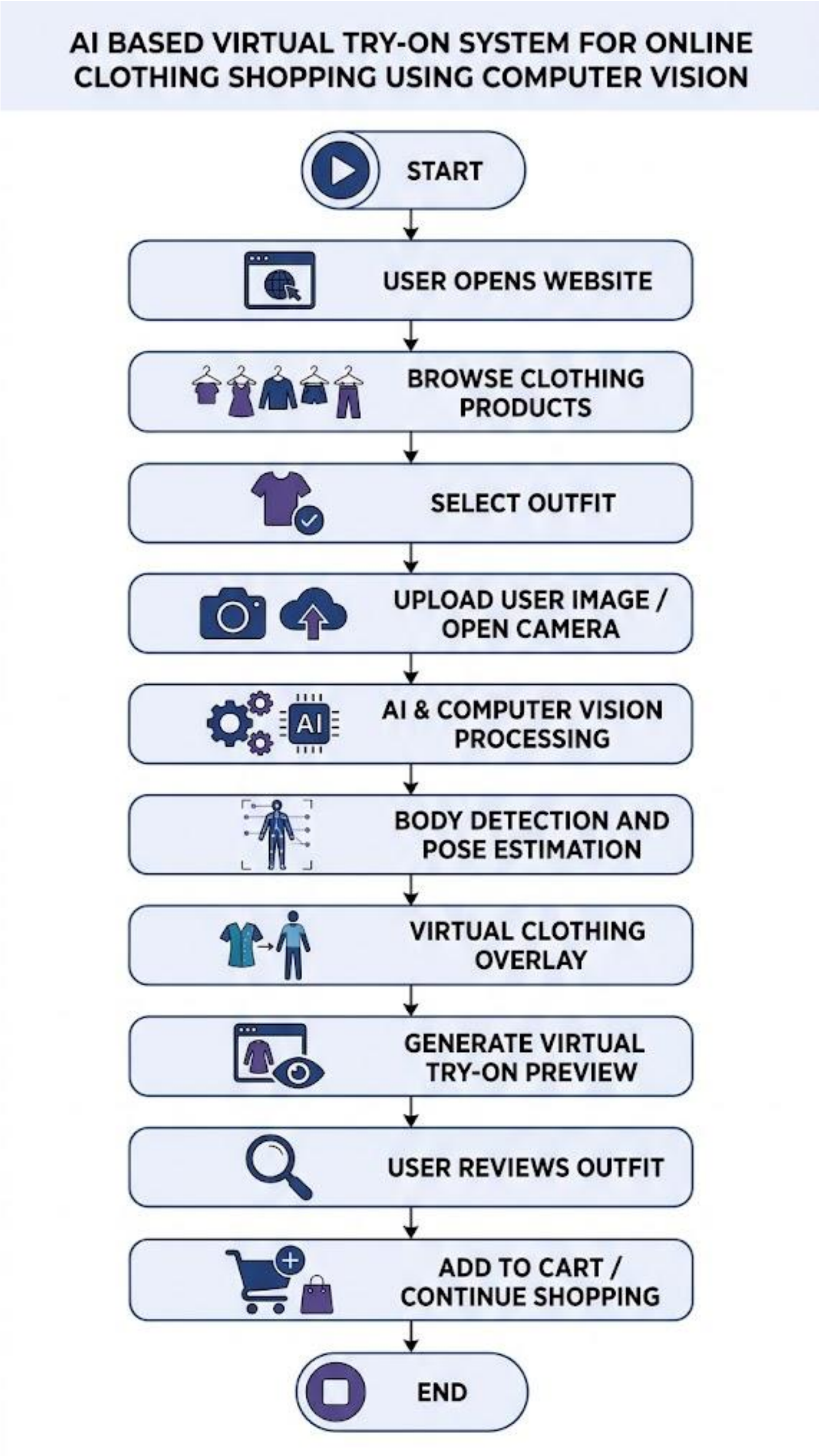
b) Define - The system was designed to provide realistic virtual try-on, easy image upload, and a responsive user experience.

c) Ideate - Different ideas such as AR mirrors and AI overlays were explored before selecting the virtual try-on solution.

d) Prototype - A prototype website with product browsing, virtual try-on, and shopping features was developed.

e) Test - User testing showed improved shopping confidence, usability, and engagement with the platform.

3.2 Methodology



3.3 Prototype Description

3.3.1 Materials / Technologies Used

a) HTML - Used for webpage structure and content design.

b) CSS- Used for styling, responsive layouts, and animations.

c) JavaScript - Used for frontend interactivity and dynamic content handling.

d) AI / Related APIs - Used for body detection, pose estimation, and virtual clothing overlay.

e) VS Code - Used as the development environment.

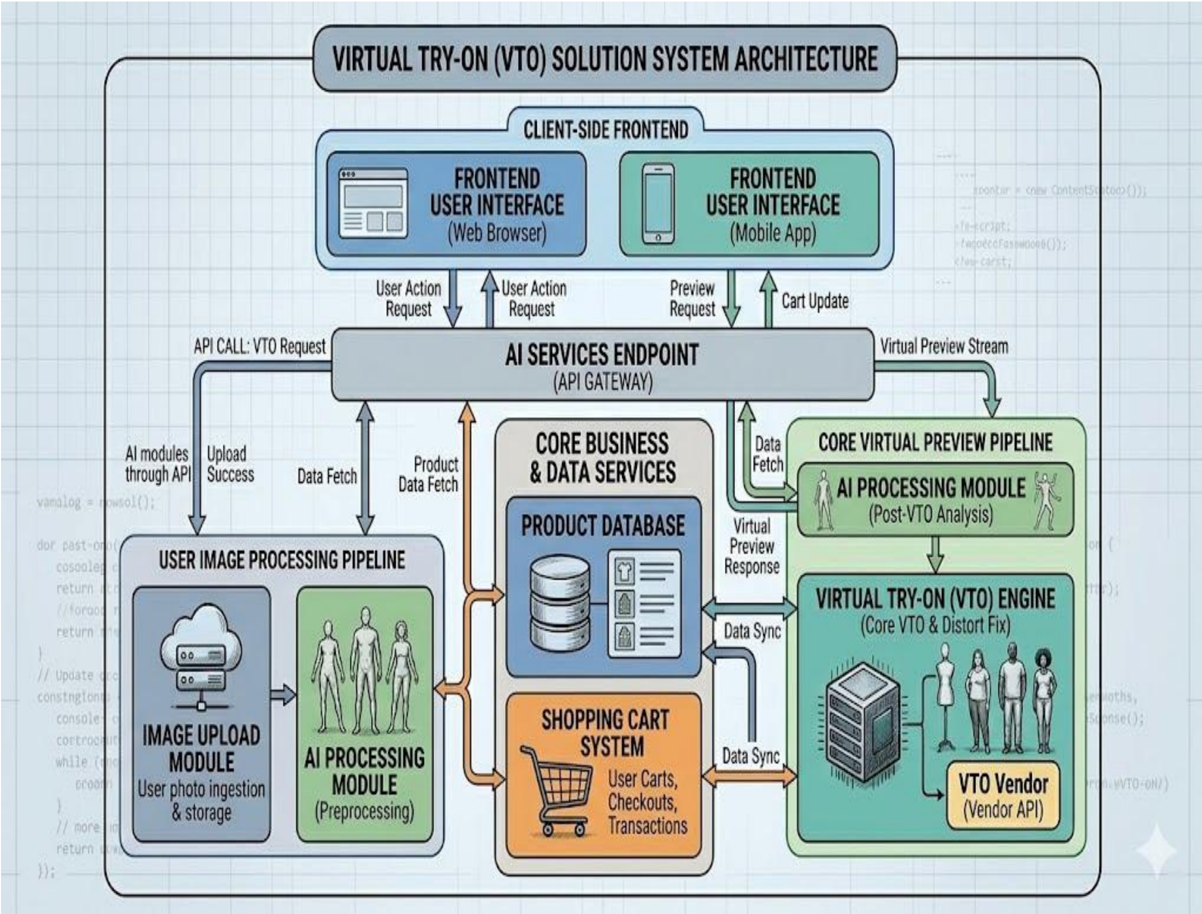
f) Browser Platform - Used for running and testing the web application.

3.3.2 System Architecture

The system architecture consists of:

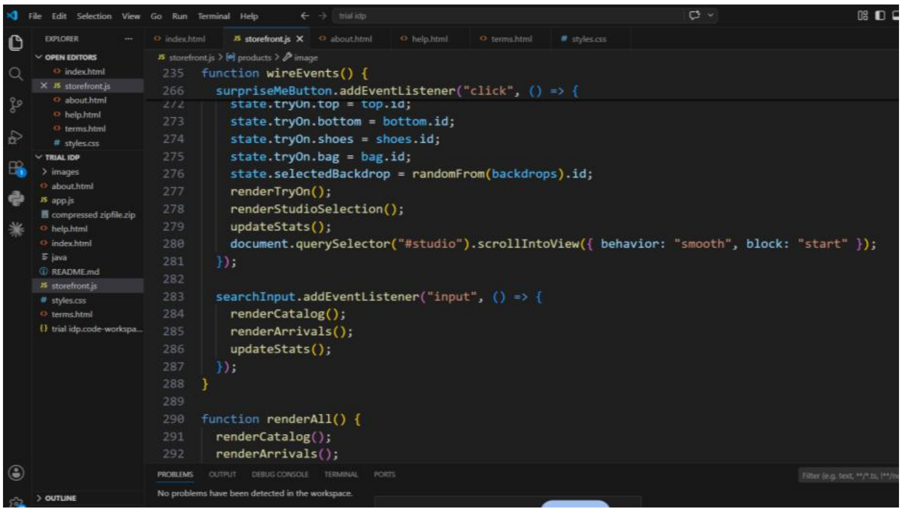
- Frontend User Interface
- Product Database
- AI Processing Module
- Image Upload Module
- Virtual Try-On Engine
- Shopping Cart System

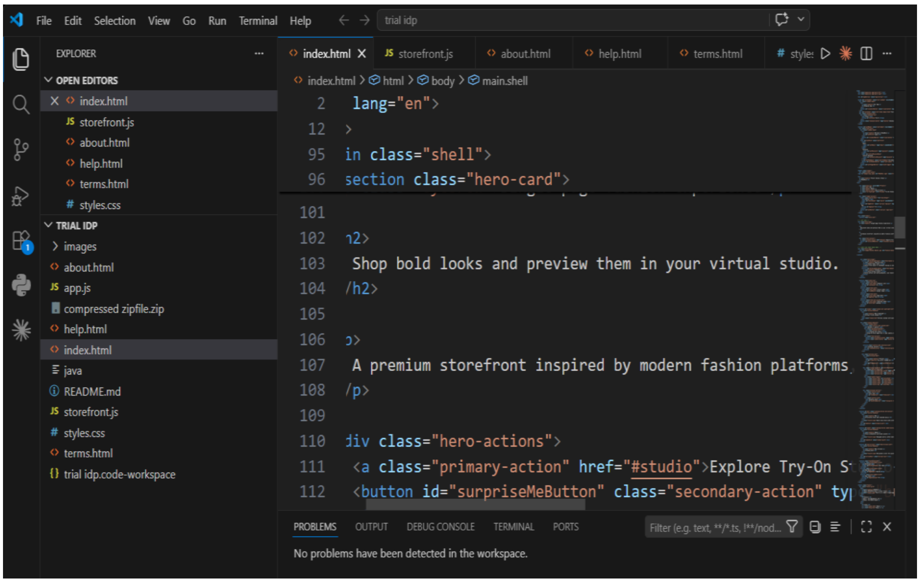
The frontend communicates with AI modules through APIs to process images and generate virtual previews.

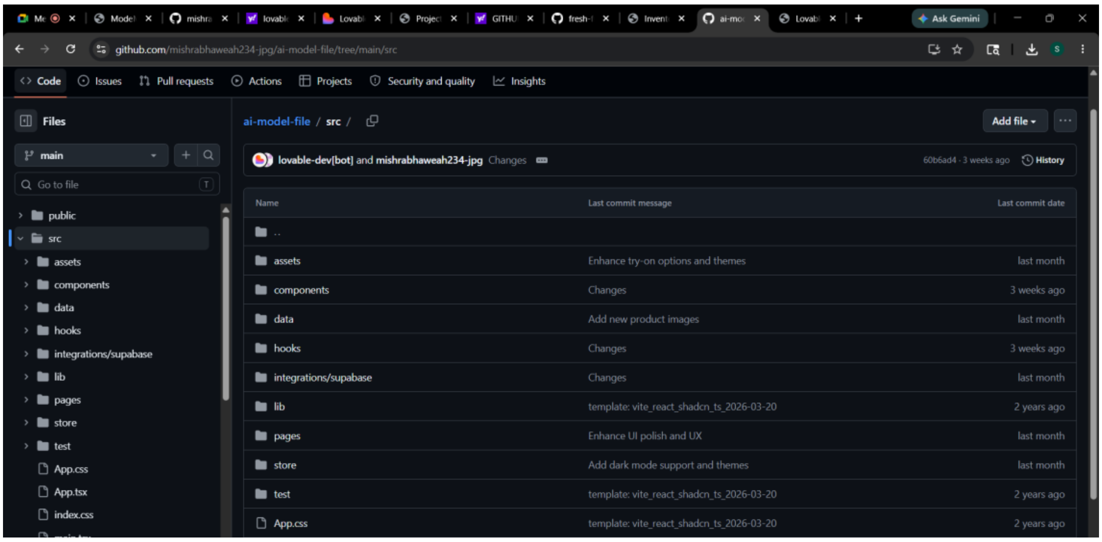


Chapter 4

Implementation



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Actions
Projects
Security and quality
Insights
Files
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Go to file
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src
assets
components
data
hooks
integrations/supabase
lib
pages
store
test
App.css
App.tsx
index.css
ai-model-file / src /
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Last commit date
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pages Enhance UI polish and UX last month
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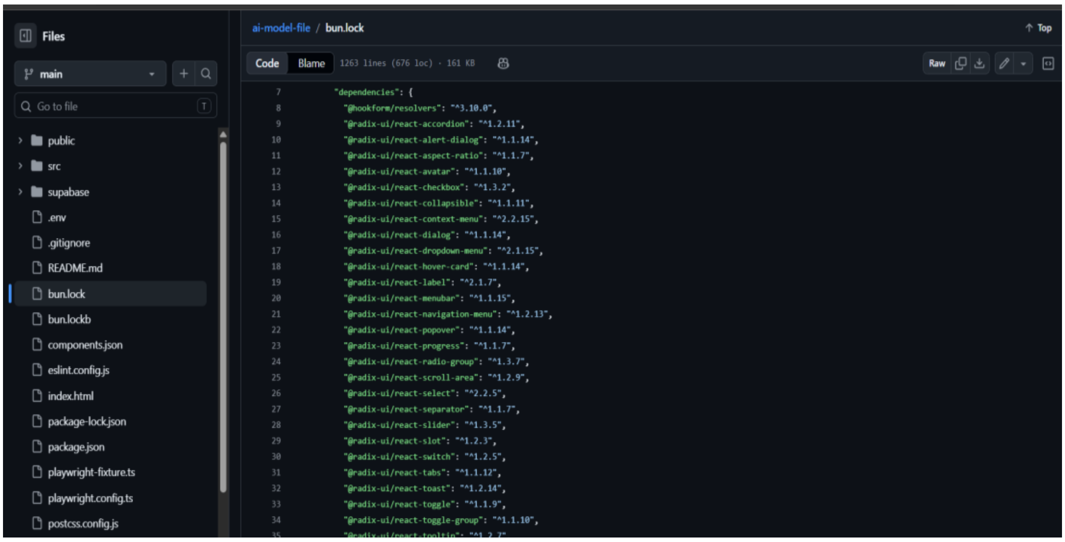
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supabase
.env
.gitignore
README.md
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Chapter 5

Results and Analysis

User Testing & Feedback

Sample

Participants: 15 students and online shoppers.

Quantitative Results

- Improved shopping visualization accuracy
- Better user engagement with interactive UI
- Faster outfit selection process
- Increased confidence before purchase
- Responsive performance across devices

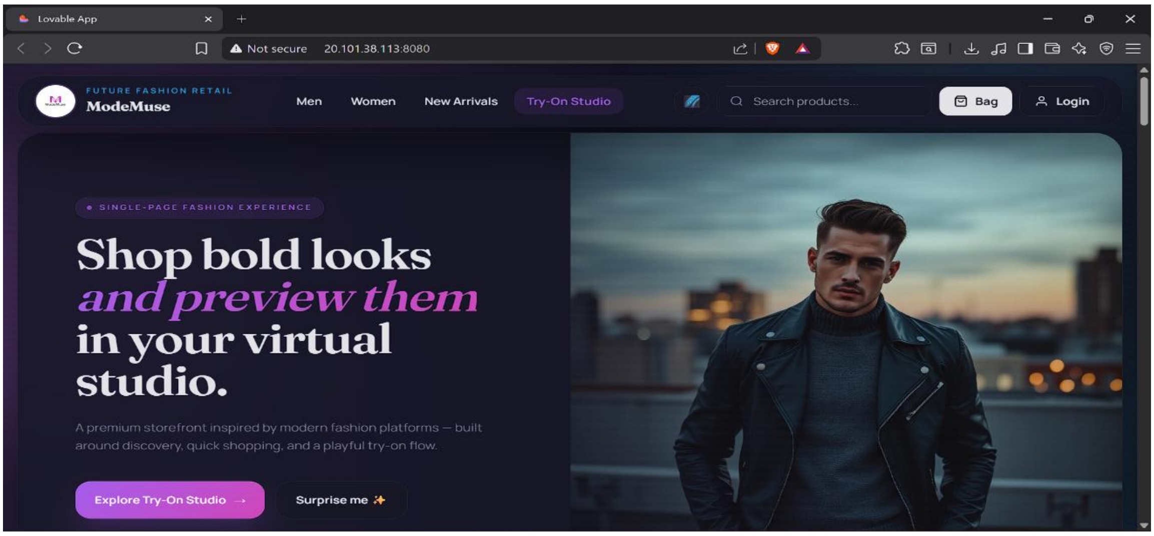
Qualitative Feedback

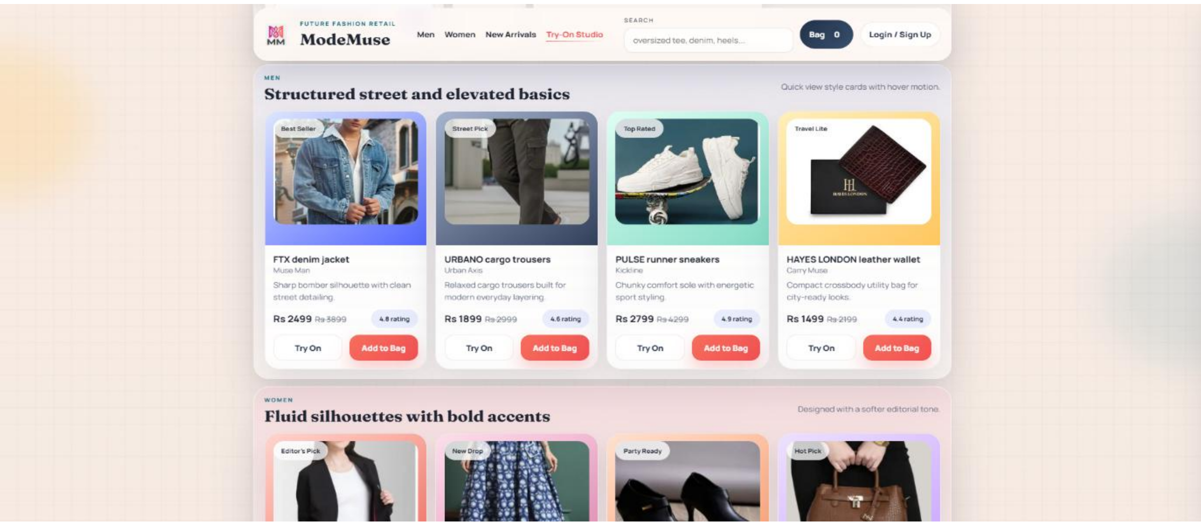
- Users: “The virtual preview makes shopping easier.”
- Testers: “The interface looks modern and attractive.”
- Reviewers: “AI try-on improves customer experience.”

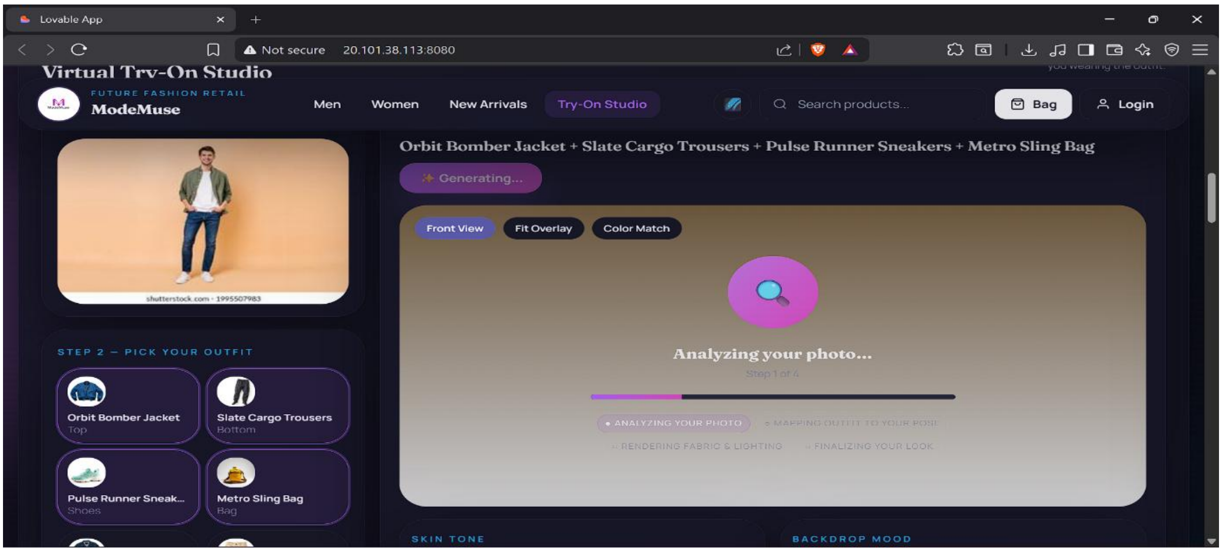
Software Output

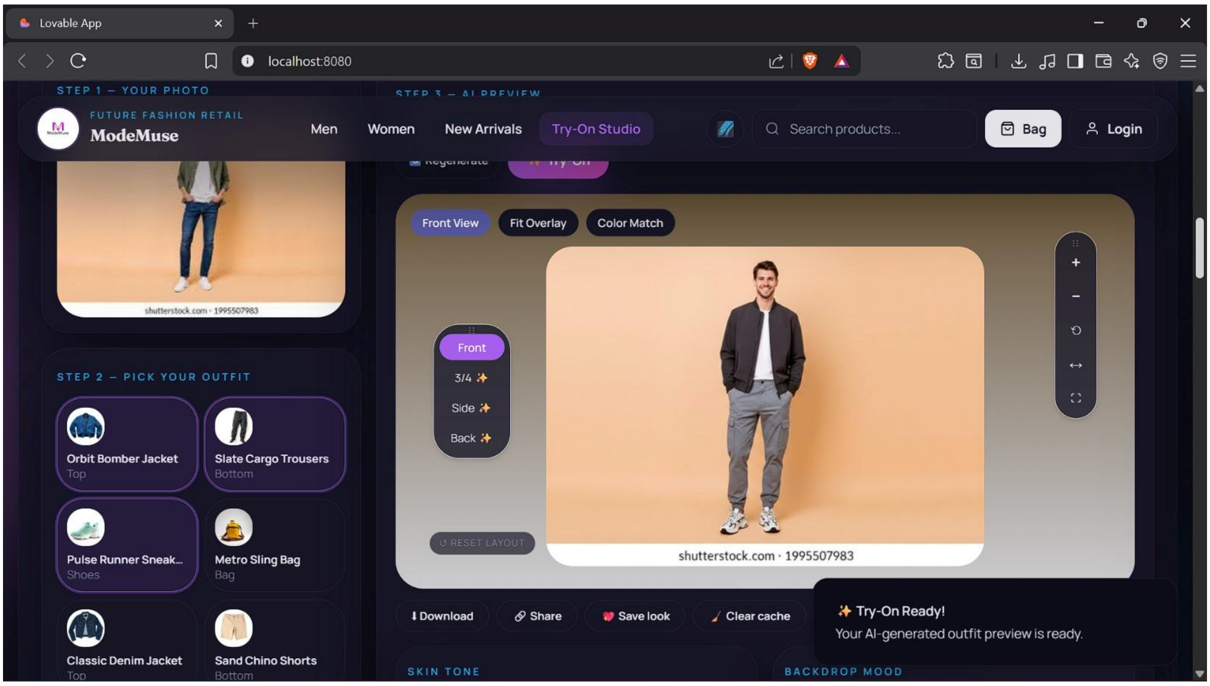
Screenshots of the developed website and virtual try-on interface are included to demonstrate system functionality and performance.

The results indicate that the project objectives were successfully achieved.






Lovable App
Not secure 20.101.38.113:8080
Virtual Try-On Studio
FUTURE FASHION RETAIL
ModeMuse
Men
Women
New Arrivals
Try-On Studio
Search products...
Bag
Login
Orbit Bomber Jacket + Slate Cargo Trousers + Pulse Runner Sneakers + Metro Sling Bag
Generating...
Front View
Fit Overlay
Color Match
Analyzing your photo...
STEP 2 — PICK YOUR OUTFIT
Orbit Bomber Jacket
Top
Slate Cargo Trousers
Bottom
Pulse Runner Sneak...
Shoes
Metro Sling Bag
Bag
SKIN TONE
BACKDROP MOOD


Lovable App
localhost:8080
STEP 1 — YOUR PHOTO
FUTURE FASHION RETAIL
ModeMuse
Men
Women
New Arrivals
Try-On Studio
Search products...
Bag
Login
Front View
Fit Overlay
Color Match
Front
3/4
Side
Back
RESET LAYOUT
shutterstock.com · 1995507983
STEP 2 — PICK YOUR OUTFIT
Orbit Bomber Jacket
Top
Slate Cargo Trousers
Bottom
Pulse Runner Sneak...
Shoes
Metro Sling Bag
Bag
Classic Denim Jacket
Top
Sand Chino Shorts
Bottom
Download
Share
Save look
Clear cache
Try-On Ready!
Your AI-generated outfit preview is ready.
SKIN TONE
BACKDROP MOOD

Chapter 6

Conclusion & Future Work

The AI Based Virtual Try-On System successfully improves the online clothing shopping experience by combining AI, and modern web technologies. The system provides users with a realistic outfit preview, helping them make better purchasing decisions and reducing uncertainty.

The project demonstrates how AI-driven virtual fitting technology can enhance customer satisfaction, improve engagement, and reduce product returns in e-commerce platforms.

Future Work

- Real-time Augmented Reality (AR) try-on
- 3D avatar generation
- AI-based fashion recommendations
- Automatic size prediction
- Multi-angle outfit visualization
- Integration with online payment systems
- Mobile application development

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Research Papers

- Virtual Try-On Network for Clothing Transfer using Computer Vision
- Deep Learning Techniques for Fashion Recommendation Systems
- AI Applications in E-Commerce and Fashion Retail

Text Books

- Digital Image Processing – Rafael C. Gonzalez
- Computer Vision and Pattern Recognition – Richard Szeliski

Online Articles

- https://en.wikipedia.org/wiki/Computer_vision
- https://en.wikipedia.org/wiki/Augmented_reality
- https://developer.mozilla.org/

ANNEXURES

Annexure A – User Feedback Forms

Annexure B – UI Design Iterations

Annexure C – Team Roles and Contributions